Branching Fidelity

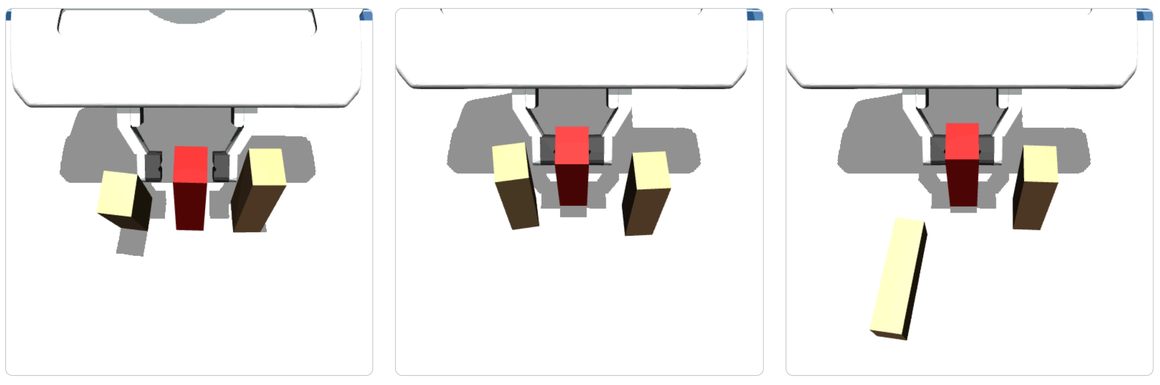
Testing World Models as Intuitive Physics Engines for Robot Safety

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Many manipulation tasks hinge on millimetres. A world model useful for them must branch where physics branches, and average prediction error does not test that. The fork probe does.

1 Small errors, different outcomes

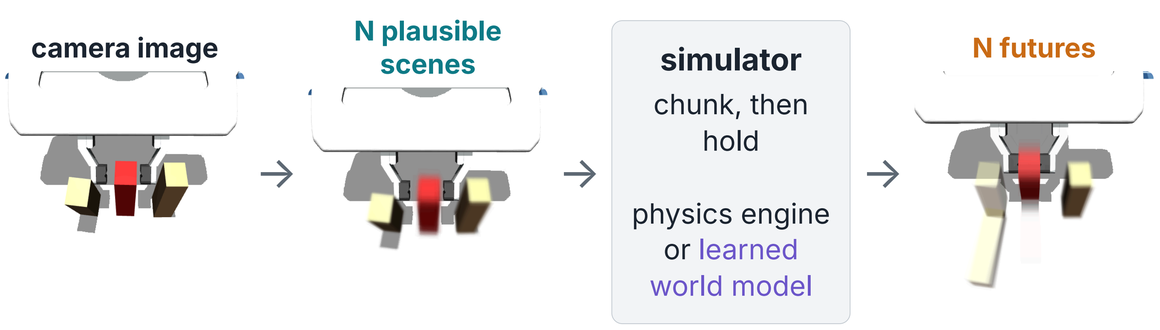
Picking from a stack, pushing a tall bottle, inserting a peg, nudging an object near an edge: **a few millimetres of error in the start or the action decide the outcome.** Here the same command lifts the middle block cleanly or tips a neighbour.



the chunk starts | half the futures: neighbour stays | the other half: neighbour topples

A state whose futures, sampled over the robot's own small errors, split into different outcomes is a **fork**. A world model's **branching fidelity**: does it fork where physics does?

2 The fork probe: a branching-fidelity test



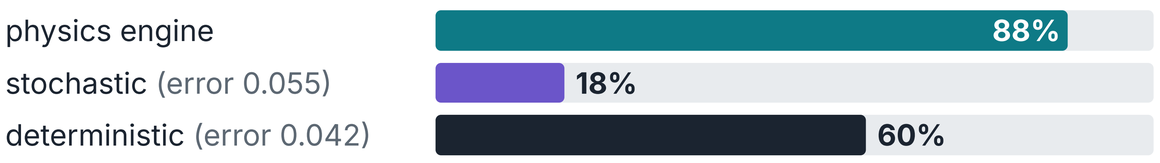
- Samples the robot's **measured** perception (2.5 mm) and execution (~3 mm) errors.
- Flags **two separate groups**, not just spread: futures can vary smoothly with one outcome; a fork has a gap.
- No task labels, failure data or tuned thresholds.

Upper reference: the physics engine. MuJoCo with the true masses and friction is not deployable, but sets the bar: **88%** of forks caught, **7%** false alarms, unseen layouts, one camera image.

3 Similar error, different branching

Two world models in DINOv2 feature space. By average error they look alike, both **3–4× better than predicting that nothing moves**. The fork probe separates them.

FORKS CAUGHT · SAME UNSEEN TEST SET



4 Why

In physics, futures differ only because starts and commands differ, and diverge only near the boundary between outcomes.

Stochastic: noise in the wrong place

Fresh noise at every step: the right *amount* of spread, but in every direction, almost as much on safe states as on forks (1.3× vs 4.8× in physics).

Deterministic: right place, no gap

Spread comes only from perturbed inputs, as in physics: along the branch, 32× larger on forks. But squared-error training predicts the average, blurring the branches.

5 Fork-aware training

Uncertain starts: put the uncertainty where it is, in the robot's knowledge of the scene; perturb the start, roll out deterministically. **Curate with the split test:** find each training scene's rare branch without labels and oversample it.

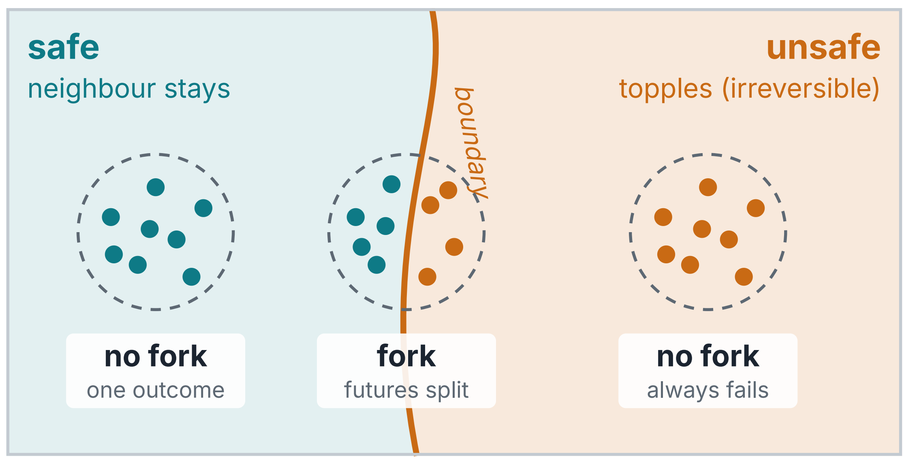
FORKS CAUGHT · TWO FRESH TEST SETS



False alarms 7–11%. Part of the gain may come from a faster training recipe. A generative model given the same uncertain starts still catches 0%.

6 Application: a label-free safety signal

People judge whether a stack will fall by **imagining it a few times** with slightly different guesses of where things are; if the outcomes disagree, it is precarious (the **intuitive physics engine** of cognitive science). The probe is this check with measured errors. When one branch is irreversible, as for toppling rigid bodies, a fork marks the **edge of the safe set**.



A proxy, not a failure detector, and not for every task: a fork with two harmless branches is a false alarm, an action that always fails never splits, and gradual harm (spills) need not split. Our follow-up builds a label-free safety filter on this signal.

7 Takeaways

Report branching fidelity
beside average error: does the model fork where physics does?

Uncertainty in the start
Deterministic dynamics from perturbed starts keep the branch; per-step noise smears it.

Train for rare branches
The probe's own split test finds them without labels.

Forks flag the safe-set edge
in rigid-body tasks; pair them with a harm signal.

Limitations: simulation, one three-block task; the ground truth uses the same error model the probe samples; small world models, trained once.